Technical Requirements Document (TRD) v2.0

CloudWise AI

Version: 2.0
Status: Active Development
Last Updated: June 2026

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1. Introduction

Purpose

This document defines the technical architecture, infrastructure requirements, system components, integrations, data flow, and engineering standards required for CloudWise AI.

It serves as the primary technical reference for development, deployment, testing, and future scaling.

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2. System Overview

CloudWise AI is a cloud infrastructure intelligence platform that connects to AWS or LocalStack environments, discovers resources, analyzes costs, generates optimization recommendations, and provides actionable insights through a modern web dashboard.

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3. Technology Stack

Frontend

Framework:

- React 19
- TypeScript
- Vite

Libraries:

- React Router
- React Query
- Recharts

- Axios

Styling:

- Vanilla CSS
- CSS Variables
- Responsive Layout System

Backend

Framework:

- FastAPI

Language:

- Python 3.12+

Key Libraries:

- SQLAlchemy
- Alembic
- Boto3
- Pydantic
- Passlib
- Python-JOSE

Database

Development:

- SQLite

Production:

- PostgreSQL

ORM:

- SQLAlchemy

Migration Tool:

- Alembic

Cloud Services

Provider:

- AWS

SDK:

- Boto3

Supported Services:

- EC2
- EBS

Planned:

- S3
- RDS
- Lambda

Local Development

LocalStack

Endpoint:

http://localhost:4566

Purpose:

- Offline AWS simulation
- Demo environment
- Development testing

Hosting

Frontend:

Vercel

Backend:

Render

Database:

Render PostgreSQL

Optional Cache:

Redis / Upstash

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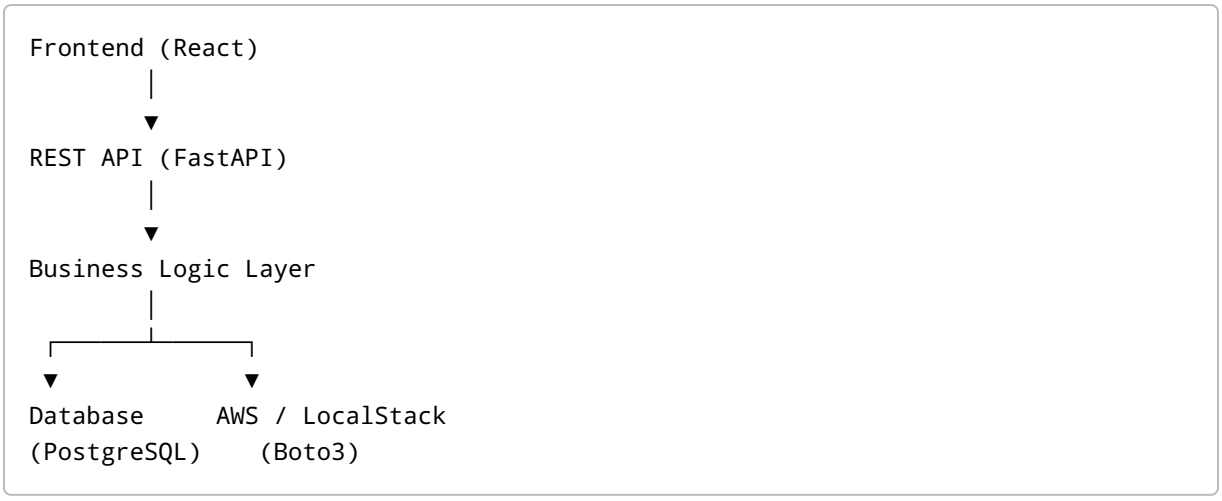
4. System Architecture

High-Level Architecture



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5. Core Modules

Authentication Module

Responsibilities:

- User Registration
- Login
- JWT Creation
- Token Validation
- Password Reset

Endpoints:

```
POST /auth/register
POST /auth/login
POST /auth/forgot-password
POST /auth/reset-password
```

Cloud Integration Module

Responsibilities:

- AWS Validation
- LocalStack Validation
- Session Creation
- Resource Discovery

Files:

```
backend/app/services/aws_service.py
```

Functions:

```
connect_aws()
discover_ec2()
discover_ebs()
get_cost_data()
get_cloudwatch_metrics()
```

Resource Discovery Module

Responsibilities:

- Discover infrastructure resources
- Normalize cloud data
- Store inventory

Supported Resources:

EC2

Collected Fields:

- Instance ID
- Instance Type
- State
- Region
- Tags

EBS

Collected Fields:

- Volume ID
- Size
- State
- Attachments
- Encryption

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6. LocalStack Architecture

Toggle Flow

Frontend

↓

use_localstack = true

↓

Backend API

↓

connect_aws()

↓

endpoint_url = http://localhost:4566

↓

LocalStack Services

Endpoint Switching Logic

AWS:

```
endpoint_url = None
```

LocalStack:

```
endpoint_url = "http://localhost:4566"
```

Applied To:

- STS
- EC2
- CloudWatch
- Cost Explorer

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7. Database Architecture

Core Tables

users

Purpose:

Authentication

Fields:

```
id
email
hashed_password
created_at
updated_at
```

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cloud_accounts

Purpose:

Store cloud connections

Fields:

```
id
user_id
account_id
provider
region
use_localstack
created_at
```

---

resource_inventory

Purpose:

Store discovered resources

Fields:

```
id
resource_id
resource_name
resource_type
region
status
arn
tags
metadata_json
created_at
```

---

recommendations

Purpose:

Store optimization suggestions

Fields:

```
id
resource_id
recommendation_type
priority
estimated_savings
description
created_at
```

---

metrics

Purpose:

Store utilization data

Fields:

```
id
resource_id
metric_name
metric_value
timestamp
```

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8. API Architecture

Authentication APIs

```
POST /auth/register
POST /auth/login
POST /auth/forgot-password
POST /auth/reset-password
```

Cloud APIs

```
POST /cloud/connect
POST /cloud/scan
GET /cloud/resources
```

Dashboard APIs

```
GET /dashboard
GET /dashboard/summary
GET /dashboard/costs
GET /dashboard/health-score
```

Recommendation APIs

```
GET /recommendations
GET /recommendations/{id}
```

Forecast APIs

```
GET /forecast
```

Report APIs

```
POST /reports/generate
GET /reports
```

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9. Recommendation Engine

EC2 Rules

Idle Instance

Condition:

```
CPU < 5%
```

Output:

```
Recommend shutdown
```

Underutilized Instance

Condition:

```
CPU < 20%
```

Output:

```
Recommend downsizing
```

EBS Rules

Unattached Volume

Condition:

```
No active attachment
```

Output:

```
Recommend deletion
```

10. Cost Analysis Engine

Current Source:

AWS Cost Explorer

Responsibilities:

- Daily cost calculation
- Monthly cost calculation
- Service-based grouping

Outputs:

```
Current Spend
Estimated Spend
Potential Savings
```

11. Dashboard Engine

Data Sources:

- Resource Inventory
- Metrics
- Recommendations
- Cost Data

KPIs:

- Total Resources
- Monthly Spend
- Potential Savings
- Health Score

Charts:

- Cost Trend
- Resource Distribution

12. Security Requirements

Authentication:

- JWT Tokens
- Password Hashing
- Protected Routes

Credential Protection:

- Never store AWS secrets in plaintext
- Encrypt sensitive values

API Security:

- Input validation
- Request sanitization
- Rate limiting (future)

Transport Security:

- HTTPS only in production

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13. Logging Requirements

Required Logs:

Authentication:

```
User Login
Failed Login
Password Reset
```

Cloud:

```
AWS Connection
LocalStack Connection
Resource Discovery
```

Infrastructure:

```
EC2 Discovery Count
EBS Discovery Count
Resources Saved
Dashboard Requests
```

Errors:

```
AWS Failures
Database Errors
API Exceptions
```

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14. Performance Requirements

Dashboard Load:

```
< 3 seconds
```

Cloud Scan:

```
< 60 seconds
```

API Response:

```
< 500ms average
```

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15. Deployment Requirements

Frontend:

Vercel

Backend:

Render

Database:

Render PostgreSQL

Environment Variables:

```
DATABASE_URL=
JWT_SECRET=
AWS_REGION=
LOCALSTACK_ENDPOINT=http://localhost:4566
DEMO_MODE=
```

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16. Future Technical Roadmap

Version 2.1

- Forecasting Engine
- PDF Reports
- Improved Health Score

Version 3.0

- AI Copilot
- S3 Discovery
- RDS Discovery
- Lambda Discovery

Version 4.0

- Azure Integration
- GCP Integration
- Multi-Cloud Architecture
- Redis Caching
- RBAC

Approval

Document Status:

Approved

Version:

2.0